

1000 mm FRP panel sections — structural test

Expandable house · options A–D as drawn, plus four engineered alternatives · 8 August 2026

Four panel sections were tested for section properties, local plate stability, sandwich skin wrinkling and service deflection. **The headline is that none of the four is decided by strength or stiffness.** Peak stress across all of them is 1.8–7.2 MPa against a 590 MPa material — about **1 % utilisation** — and deflections run L/637 to L/2404 where codes ask L/240. The four differ in weight and in local stability, and in nothing else that matters.

LIGHTEST OF YOUR FOUR

D — 13.19 kg/m²

3.90 kg/m² under B

STRONGEST IN BENDING

B — Zx 258 cm³

29 % more capacity than D — and unused

STRONGEST LOCALLY

C — 690 MPa

edge-wall buckling, 3.2× B at 21 % less weight

LIGHTEST OVERALL

H — 10.56 kg/m²

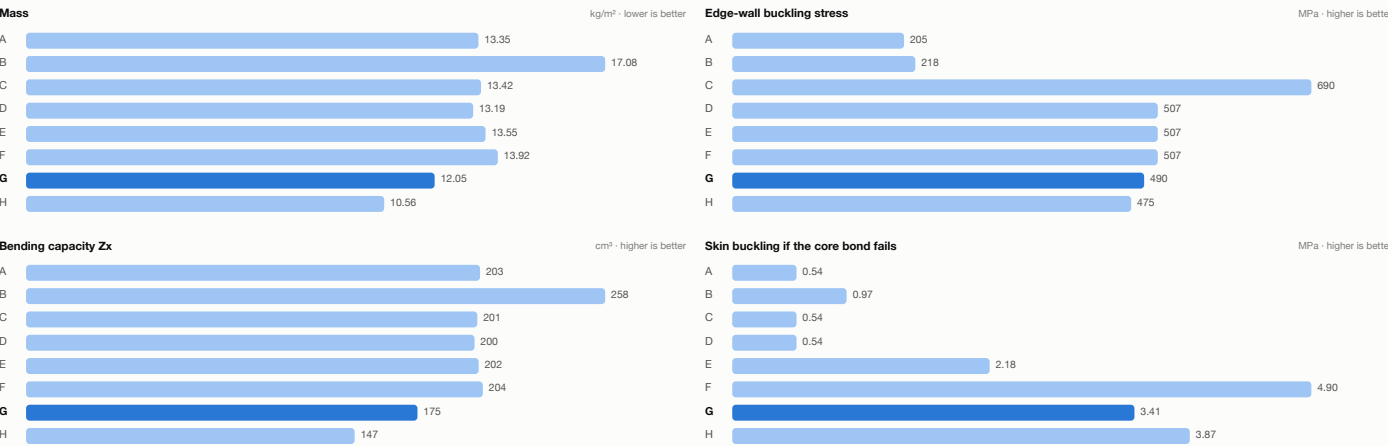
428 kg saved over D across the house

Which is lightest, which is strongest. Of the four you drew: **D is lightest** at 13.19 kg/m² and **B is strongest in bending** at Zx 258 cm³. But "strongest" is the wrong prize here — B's extra capacity is 29 % more of something already eighty times oversupplied, bought with 29 % more material. The strength that *is* scarce is local: **C is strongest against buckling** at 690 MPa, beating B by 3.2× while weighing 21 % less. Across everything tested, **H is lightest overall** at 10.56 kg/m².

Why weight is the only real currency

Stiffness per kilogram is **identical across all four** — 22.2 to 22.5 kNm² per kg/m². The 3–4 mm skins sitting at ±35 mm from the neutral axis produce essentially all of the bending stiffness; the 35 mm edge profiles are only 3.5 % of a 1000 mm module and contribute almost nothing. So extra material buys exactly proportional stiffness and **no efficiency gain at all**. This matches the 15 June finding that skins carry 89–90 % of panel bending stiffness.

The consequence is direct: **B costs 3.90 kg/m² more than D**. Over roughly 163 m² of panel in the 45 m² unit — 73 m² of wall, 45 m² of floor, 45 m² of roof — that is **about 635 kg of added mass** on a folding house whose entire value proposition is transport weight.



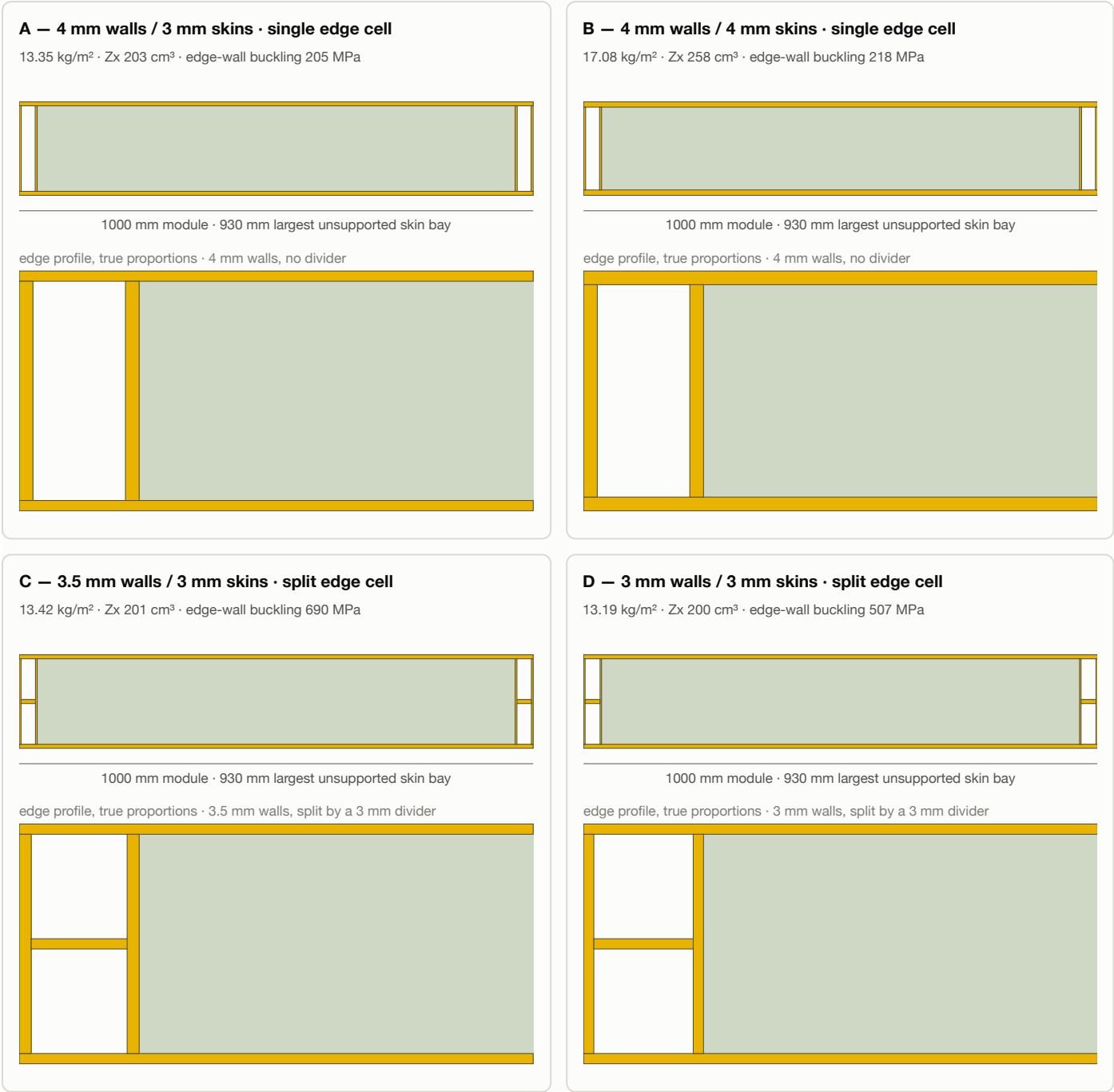
Full results

Option	Skins	Walls	Mid webs	Mass kg/m ²	Ix cm ⁴	Zx cm ³	EI kNm ²	Edge-wall buckling MPa	Bare skin buckling MPa	Wall defl.	Floor defl.
A	3 mm	4 mm	0	13.35	709	203	301	205	0.54	L/1975	L/680
B	4 mm	4 mm	0	17.08	904	258	383	218	0.97	L/2404	L/819
C	3 mm	3.5 mm	0	13.42	704	201	299	690	0.54	L/1927	L/660
D	3 mm	3 mm	0	13.19	700	200	297	507	0.54	L/1869	L/637
E	3 mm	3 mm	1	13.55	707	202	300	507	2.18	L/2002	L/691
F	3 mm	3 mm	2	13.92	713	204	302	507	4.90	L/2087	L/726
G	2.5 mm	3 mm	2	12.05	611	175	259	490	3.41	L/1819	L/636
H	2 mm	3 mm	3	10.56	513	147	217	475	3.87	L/1576	L/555

Wall case 2.706 m span at 0.5 kPa wind; floor case 2.4 m span at 2.0 kPa live load. Deflections include core shear. Rows F–H are the engineered alternatives, not part of the original drawing.

The four sections as drawn

Yellow is FRP, green is the bonded stone-wool core. Depth is drawn x2.6 so the 3 mm walls are visible; the 1000 mm width is to scale.



What actually governs this panel — the bond, not the section

The unsupported skin bay between the edge webs is **930 mm** in every one of A–D. With the core bonded, the governing mode is skin wrinkling at **40.5 MPa**; against the 7.2 MPa floor-panel stress that is a factor of **5.6** — by far the lowest margin anywhere in this analysis and the only one within an order of magnitude of the applied load. If that bond ever fails, the bare 930 mm skin buckles at **0.54 MPa**, which is effectively zero.

This is where the money should go. Core density, adhesive coverage, flatwise tensile and climbing-drum peel testing govern this panel. No choice between A, B, C and D changes the number above. Every one of the three alternatives below attacks it directly, by shortening the bay.

Three better options to choose from

All three keep the 70 mm depth, the split edge cells and the 3 mm edge walls. They differ in skin thickness and in how many intermediate webs brace the skin.

F — Safe — nothing new to buy

3 mm skins · 2 intermediate webs · 13.92 kg/m² (+6 % vs D) · bare-skin buckling 4.90 MPa (9.0× D) · wall L/2087



1000 mm module · 310 mm largest unsupported skin bay

Keeps the 3 mm sheet you have already tested, so there is no new material qualification and no supplier risk. Two intermediate webs cut the unsupported skin bay from 930 mm to 310 mm, which raises bare-skin buckling nine times over D and gives the stiffest, least-deflecting panel of the whole 3 mm family. Costs 0.73 kg/m² more than D.

G — Balanced — the one I would build recommended

2.5 mm skins · 2 intermediate webs · 12.05 kg/m² (-9 % vs D) · bare-skin buckling 3.41 MPa (6.3× D) · wall L/1819

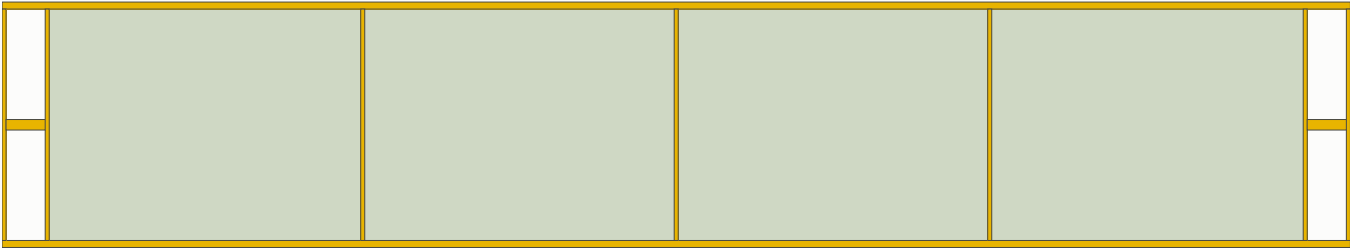


1000 mm module · 310 mm largest unsupported skin bay

Drops the skin to 2.5 mm and pays for it with web bracing. That trade is favourable because the panel has roughly eighty times more bending capacity than it needs, and skins are 86 % of the material. Result: lighter than every option you drew, still six times better braced than D, and deflection essentially unchanged at L/1819.

H — Light — maximum weight saving

2 mm skins · 3 intermediate webs · 10.56 kg/m² (-20 % vs D) · bare-skin buckling 3.87 MPa (7.1× D) · wall L/1576



1000 mm module · 232 mm largest unsupported skin bay

2 mm skins with three intermediate webs. 20 % lighter than D and 36 % lighter than B — about 428 kg saved over the house against D. Bending stress rises to 9.8 MPa, still under 2 % utilisation. The cost is real but not structural: a 2 mm sheet your supplier has not proven, and less margin against impact damage and screw bearing.

How to choose. Take **F** if you want zero supplier risk — it uses only the 3 mm sheet already tested and simply braces it properly. Take **G** if you want the best overall balance: it is lighter than anything you drew, six times better braced than D, and saves about 185 kg over the house. Take **H** only if weight is the binding constraint and your supplier will commit to a qualified 2 mm sheet — it saves about 428 kg over D but gives up impact and screw-bearing margin. **My recommendation is G.**

Recommendation

1. **Do not build B.** It is the heaviest by 29 % and buys only stiffness that is already surplus — roughly 635 kg for nothing.

2. **If you must choose from the drawing, choose D** — lightest, and its split edge cell gives 2.5× the local buckling strength of A. Choose C instead only if the supplier cannot hold 3.0 mm walls consistently; C costs just 0.23 kg/m² more and has the best local number of the four.
3. **Better: build G** — 2.5 mm skins with two intermediate webs. 12.05 kg/m², lighter than every drawn option, 310 mm skin bays instead of 930 mm, deflection unchanged in practice.
4. **Spend the engineering budget on the skin-to-core bond**, not on section geometry. That is the only place in this panel where the margin is thin.

Assumptions and what still needs confirming

Assumed, because the drawing does not give them: section depth **70 mm** and edge-profile width **35 mm**, both taken from the established project geometry; a bonded 100 kg/m³ stone-wool core (E 5 MPa, G 2.5 MPa). Absolute numbers scale with depth — roughly with depth² for stiffness — but the ranking does not change.

- **Confirm the section depth.** This is the one input that moves every number.
- Confirm the edge-profile width. The drawing's proportions suggest end cells nearer 100 mm than the assumed 35 mm, which would change the local figures.
- Transverse modulus and shear modulus are still estimated at 10 and 4 GPa — supplier test data would firm up every buckling number here.
- Get the real core data sheet. Since wrinkling governs, E and G of the core matter more than anything about the FRP.
- Run flatwise tensile and climbing-drum peel tests on the skin/core bond.

Material: tested pultruded E-glass/epoxy, E_L 42.4 GPa, tensile 590 MPa, density 1.90 g/cm³ (report DZTC/BG-24092301). Preliminary analytical section test — elastic section properties, orthotropic local plate buckling (Kollár-type closed form), Hoff/Plantema skin wrinkling, and simply supported service deflection including core shear. This is not FEA and not a certification calculation. Source: analysis/panel_section_tests_2026_08_08.py · Interactive version: safariibg.github.io/expandable-house-viz/panel-section-tests.html